



Customer Shopping Behavior Analysis

Insights from **3,900 transactions** across product categories

PYTHON

POSTGRESQL

POWER BI

Project Overview

01

Data Collection

3,900 purchase records · 18 features including demographics, purchase details & behavior

03

SQL Analysis

10 business queries in PostgreSQL covering revenue, segments & discount patterns

02

Python EDA

Data cleaning, feature engineering, median imputation for 37 missing review ratings

04

Power BI Dashboard

Interactive visuals with filters by subscription, gender & category



Revenue Insights

2.1x

Male vs. Female Revenue

Male customers generate over twice the revenue of female customers

\$62K

Young Adults Lead

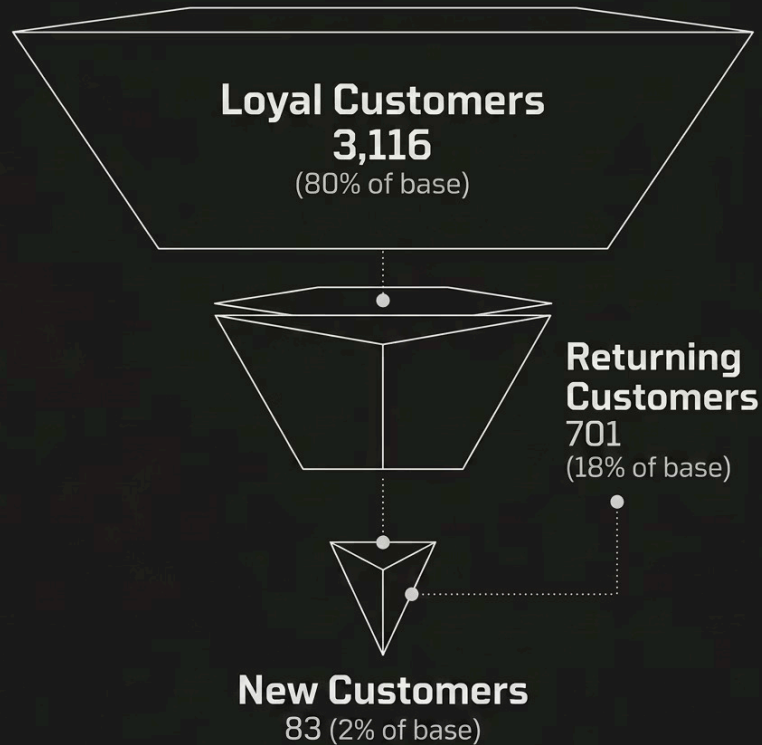
Highest revenue of any age group at \$62,143 total spend

3,900

Total Transactions

Across all product categories and customer segments

Customer Segmentation



Loyal Customers — 3,116

80% of base · Strong retention foundation

Returning Customers — 701

18% of base · Active re-engagement segment

New Customers — 83

2% only · Acquisition pipeline critically weak

⚠ Only 2% new customers — strong retention but weak acquisition pipeline needs urgent attention.

Subscription Analysis

Metric	Subscribers	Non-Subscribers
Total Customers	1,053	2,847
Avg Spend / txn	\$59.49	\$59.87
Total Revenue	\$62,645	\$170,436
Repeat Buyers (>5 purchases)	958	2,518

Key Insight

Average spend is **nearly identical** across both groups. Subscription value lies in **long-term retention**, not higher order value.

Upsell on **loyalty perks** — not discounts.

- ① 73% of customers are unsubscribed — a major opportunity to grow recurring revenue.

Discount & Product Analysis

Top 5 Products by Rating

Rank	Product	Avg Rating
1	Gloves	★ 3.86
2	Sandals	★ 3.84
3	Boots	★ 3.82
4	Hat	★ 3.80
5	Skirt	★ 3.78



⚠ Hat & Sneakers carry ~50% discount rates — customers may be conditioned to wait for sales, eroding margins.

Shipping & Category Insights

Standard Shipping

Avg purchase: **\$58.46**

Express Shipping

Avg purchase: **\$60.48**

839 customers used discounts but still spent **above average** — discount sensitivity is manageable.

Top 3 Products per Category

Category	#1	#2	#3
Accessories	Jewelry (171)	Sunglasses (161)	Belt (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	Vest (154)

Business Recommendations

1

Boost Subscriptions

73% unsubscribed. Promote loyalty perks — early access, free shipping — to convert returning buyers.

2

Fix Acquisition Funnel

Only 83 new customers (2%). Target campaigns toward Young Adults — the highest-revenue age group.

3

Review Discount Policy

Hat & Sneakers at ~50% discount rate. Introduce time-limited or loyalty-gated offers to protect margins.

4

Promote Top-Rated Products

Gloves, Sandals, Boots lead in ratings but not in sales. Feature in campaigns to drive higher-value purchases.

5

Upsell Express Shipping

Express users spend \$2 more per order. Bundle with subscriptions or order thresholds to increase AOV.

Thank You

Explore the full project and code on GitHub

[View on GitHub](#)

Python

EDA & Feature Engineering

PostgreSQL

10 Business Queries

Power BI

Interactive Dashboard